# Vectors Quiz, Version A



NAME DATE PERIOD SCORE / 16

Show your work in the boxes and box your final answers with units. Round calculations to two significant figures. Angles on the 0–360° scale start at east and run counterclockwise: north is 90°, west is 180°, south is 270°. A negative angle runs clockwise from east. Any correct way of writing a direction counts.

**Equations you may use**

$R_x = R\cos\theta$ $R_y = R\sin\theta$ $\tan\theta = R_y / R_x$

$R^2 = R_x^2 + R_y^2$ $R_x = A_x + B_x$ $R_y = A_y + B_y$

**1.** 2 pts

Restate 215° in two other ways.

**2.** 2 pts

Restate 40° north of west in two other ways.

**3.** 2 pts

Restate −70° in two other ways.

**4.** 5 pts

Pac-Man travels 45 m north and then 28 m west through the maze. Find his resultant displacement: magnitude and direction.

**5.** 5 pts

Sam Wilson flies the Falcon at 240 km/h at 40° north of east. A wind blowing toward 30° south of east at 65 km/h also acts on him. Find his resultant velocity: magnitude and direction.